

# Machine Learning from First Principles

B04 - Exact 100-page production blueprint

Audience: Learners who know Python and core AI mathematics

# Page architecture

| Pages  | Section                                                     |
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# Chapter requirements

## Chapter 1: The ML Workflow and Honest Baselines

Open with measurable objectives and prerequisites. Build intuition before formalism. Include at least two worked examples, an accessible visual or table, progressive practice, common mistakes, independent exercises, and retrieval review. Connect to executable Python or an interactive lab where appropriate.

## Chapter 2: Regression

Open with measurable objectives and prerequisites. Build intuition before formalism. Include at least two worked examples, an accessible visual or table, progressive practice, common mistakes, independent exercises, and retrieval review. Connect to executable Python or an interactive lab where appropriate.

## Chapter 3: Classification and Logistic Regression

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## Chapter 4: Distance-Based and Probabilistic Models

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## Chapter 5: Trees Forests and Ensembles

Open with measurable objectives and prerequisites. Build intuition before formalism. Include at least two worked examples, an accessible visual or table, progressive practice, common mistakes, independent exercises, and retrieval review. Connect to executable Python or an interactive lab where appropriate.

## Chapter requirements - continued

### Chapter 6: Support Vector Machines

Open with measurable objectives and prerequisites. Build intuition before formalism. Include at least two worked examples, an accessible visual or table, progressive practice, common mistakes, independent exercises, and retrieval review. Connect to executable Python or an interactive lab where appropriate.

### Chapter 7: Clustering and Dimensionality Reduction

Open with measurable objectives and prerequisites. Build intuition before formalism. Include at least two worked examples, an accessible visual or table, progressive practice, common mistakes, independent exercises, and retrieval review. Connect to executable Python or an interactive lab where appropriate.

### Chapter 8: Evaluation Validation and Imbalance

Open with measurable objectives and prerequisites. Build intuition before formalism. Include at least two worked examples, an accessible visual or table, progressive practice, common mistakes, independent exercises, and retrieval review. Connect to executable Python or an interactive lab where appropriate.

### Chapter 9: Feature Engineering Tuning and Interpretation

Open with measurable objectives and prerequisites. Build intuition before formalism. Include at least two worked examples, an accessible visual or table, progressive practice, common mistakes, independent exercises, and retrieval review. Connect to executable Python or an interactive lab where appropriate.

### Chapter 10: End-to-End Reproducible ML Project

Open with measurable objectives and prerequisites. Build intuition before formalism. Include at least two worked examples, an accessible visual or table, progressive practice, common mistakes, independent exercises, and retrieval review. Connect to executable Python or an interactive lab where appropriate.

## Capstone

Train, evaluate, interpret, and document a leakage-safe tabular ML system.

## Production and QA

Author original English content. Verify code, formulas, solutions, citations, diagrams, accessibility, and objective alignment. Render and reflow until the final book contains exactly 100 content pages without blank pages, repeated filler, oversized headings, or artificial spacing. Deliver source Markdown, plain text, PDF, answer guidance, and a QA report.